Adam E. Holbrook

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Operations engineer with 4+ years building dashboards, reporting pipelines, & automation for global payment infrastructure at Mastercard. Experienced in translating complex data into stakeholder-ready KPIs, redesigning workflows for speed & self-service, & designing alert & response protocols across distributed systems.

Professional Experience

Senior Site Reliability Engineer Mastercard | 2023 – 2025

Senior SRE for global payment infrastructure processing billions in transactions annually across 40+ countries.

- *Architected & deployed 200+ observability monitors across Splunk, Dynatrace, Domo, & Omnibus, achieving 99.99% uptime for critical payment systems.*
- *Engineered monitoring-as-code framework with automated stage testing environments, enabling developers to validate alerts before production deployment.*
- *Built secure Infrastructure-as-Code & CI/CD pipelines using Chef, Jenkins, XLR, Bitbucket, & Groovy, reducing deployment time throughout the delivery lifecycle.*
- *Led comprehensive Domo operational intelligence platform integrating disparate data sources (Salesforce, Splunk, Dynatrace, Jira, Remedy) into live dashboards for ticket volume, remediation metrics, & SLA performance across L2/L3 operations teams.*
- *Rebuilt enterprise data pipelines in Domo; reduced report generation from 2+ hours to seconds while delivering SLO/SLI metrics & BI capabilities for C-suite operations.*

Site Reliability Engineer II Mastercard | 2022 – 2023

Primary on-call SRE for MDES tokenization platform; maintained high availability through proactive monitoring, automated remediation, & rapid incident response.

- *Engineered custom unified Python-based terminal platform for Tomcat monitoring, disk management, log rotation, & service control with integrated API healthcheck capabilities; rolling dashboard accessible across all systems instead of rote commands on each host.*
- *Built companion VSCode automation toolkit with pre-configured connection details, script hotkeys, direct Splunk/Dynatrace integration, & database actions; greatly reduced release execution times & emergency response delays through streamlined operational workflows.*
- *Developed Python & Bash automation scripts for deployment validation, health checks, & compliance reporting, drastically reducing manual operational toil.*
- *Developed automation reducing manual operational work; created documentation supporting consistent, repeatable workflows across the team.*
- *Built operational tools streamlining service disruption response; trained team members on processes that improved response consistency.*

Site Reliability Engineer I Mastercard | 2021 – 2022

Supported operational reporting & incident response for distributed payment systems.

- *Built observability dashboards in Splunk & Dynatrace for real-time system health visibility.*
- *Resolved 1,000+ L2/L3 production incidents annually for distributed payment systems.*

DevOps Assistantship SIU Edwardsville / Illinois Department of Transportation | 2020 – 2021

Graduate assistantship supporting infrastructure & delivery workflows for public-sector systems.

Associate Consultant SIU Applied Research Consultants | 2019 – 2020

Delivered end-to-end survey research & evaluation consulting for local & regional stakeholders.

- *Designed & executed end-to-end survey projects for multiple regional clients: stakeholder intake, survey design in Qualtrics & SurveyMonkey, data analysis in R, Python, SQL, & SPSS, & final reporting.*
- *Built R-based descriptive analysis & visualization pipelines for reports including descriptive statistics, regression analyses, & data visualizations.*
- *Maintained an interdisciplinary statistics education platform helping students across fields access data processing resources.*

Research Assistant SIU Integrative Neuroscience Lab | 2019 – 2020

Implemented EEG data processing pipelines on supercomputing infrastructure & supported academic research.

- *Implemented MatLab & Octave EEG data processing pipelines to run in Linux environments on university supercomputers.*
- *Configured 3D imaging software for construction of high-precision electroencephalographic head models.*

Curated Skills

Program Evaluation & Research Program Evaluation, Needs Assessment, Survey Design, Survey Methodology, Research Design, Reporting & Recommendations

Data, Reporting & Analysis Python, R, Excel, SQL, Tableau, Domo, Jupyter, KPI Design

Systems & Automation Dashboards & Monitoring (Splunk, Dynatrace, Grafana, Prometheus), Process Documentation & Training, Jira, Git, Docker

Observability Grafana, Prometheus, Splunk, Dynatrace, Domo, Omnibus, Datadog

CI/CD & Automation Jenkins, GitLab CI, XLR, Bitbucket, Git, Chef, GitOps

Development Python, C++, PowerShell, Bash, Java, Spring, Groovy, R, REST/SOAP APIs

Incident Management 24/7 On-Call, Incident Postmortems, SLO/SLI, Paging Software, Alerting

Documentation Atlassian Suite, Confluence, Jira, Markdown, Agile, Chef Run Books

Databases PostgreSQL, MySQL, Oracle, MongoDB, Kafka, RabbitMQ, Redis, IBMMQ, Elasticsearch

Projects

- ***Proxmox Homelab*** *– Terraform/Ansible IaC workflows & GitLab CI/CD pipelines with separate production & staging environments with physical isolation. Containerized with Docker Compose; networking using OPNsense.*
- ***instant_resume*** *– Automated LaTeX-based resume generation pipeline with YAML single source of truth, profile-based content filtering, & Next.js web frontend with role-targeted views.*
- ***net4nets (EEG Deep Learning Classification Pipeline)*** *– Convolutional neural net achieving 94% accuracy classifying smokers from resting-state EEGs; end-to-end pipeline with time-frequency/PSD features & CNN validation workflows.*

Education

Graduate Coursework, Computer Science, SIU Edwardsville Data Science Focus | 2020 – 2021

- *Graduate assistantship focused on CI/CD & data pipeline development with Illinois DOT.*

M.S. Applied Psychology, Southern Illinois University Applied Statistics Focus | 2019 – 2020

- *Statistics & applied research focus; consulting & statistics coursework. 3.75 GPA.*
- *President, SIU Statistics Club. Graduate Advisor, SIU Photogenesis. Chair, 4+1 Program Committee.*

B.A. Psychology & Photography, Southern Illinois University Neuroscience Specialization; Documentary Specialization | 2014 – 2019

- *Neuroscience specializations: Addiction Neurophysiology; Neurotrauma & Rehabilitation.*
- *President, SIU Photogenesis. Recipient of Photography Merit Scholarships. Awards in Lux National Photography Journal & Little Muddy Film & Photography Festival.*